LivenessAuth AI

A Lightweight Offline Facial Recognition and Liveness Detection System for Zero-Network Environments

Submitted for :- NHAI Innovation Hackathon 7.0
Submitted By :- Anil Kushwaha, Kumaranil48309@gmail.com

1. Executive Summary

LivenessAuth AI is a secure, lightweight, and fully offline facial authentication solution designed specifically for field operations in remote and low-connectivity environments. The proposed system leverages edge artificial intelligence to perform facial recognition and liveness detection directly on mobile devices without requiring internet access.

The solution addresses the operational challenges faced by organizations working in remote locations where traditional cloud-based authentication systems become unreliable due to poor network connectivity.

By combining offline facial recognition, anti-spoofing mechanisms, secure local storage, and intelligent synchronization capabilities, LivenessAuth AI ensures uninterrupted, secure, and efficient identity verification.

The proposed system is optimized for Android and iOS platforms and can be seamlessly integrated into the existing Datalake 3.0 ecosystem while maintaining a lightweight model footprint and high processing speed.

2. Problem Statement

Reliable identity verification is critical for field personnel operating in remote locations. However, existing authentication systems primarily depend on cloud connectivity, making them unsuitable for regions with limited or no internet access.

In many operational scenarios, field staff are required to authenticate themselves in areas where network connectivity is unavailable. This often leads to authentication failures, operational delays, security vulnerabilities, and inaccurate attendance or activity records.

The challenge is to develop a highly accurate, lightweight, and entirely offline facial recognition and liveness detection system that can securely authenticate personnel on standard mobile devices while preventing spoofing attempts through photographs, videos, or digital screens.
The solution must operate efficiently under varying environmental conditions and integrate seamlessly with existing infrastructure.

3. Objective

The primary objective of LivenessAuth AI is to develop a secure mobile-based authentication system capable of performing offline facial recognition and liveness verification in zero-network environments.

The solution aims to :-

• Enable reliable offline authentication.
• Prevent identity fraud through robust liveness detection.
• Maintain a lightweight AI model footprint below the specified size constraints.
• Deliver authentication results in less than one second.
• Support both Android and iOS devices.
• Synchronize authentication records with cloud infrastructure once network connectivity is restored.
• Ensure seamless integration with Datalake 3.0 architecture.

4. Proposed Solution

LivenessAuth AI utilizes edge-AI technology to perform all authentication-related operations directly on the mobile device. During authentication, the device camera captures the user's face and verifies liveness through a series of anti-spoofing checks.

Upon successful liveness verification, facial embeddings are generated using a lightweight facial recognition model. The generated embeddings are compared against securely stored local templates to determine identity verification.

Authentication logs are stored locally in an encrypted database. Once network connectivity becomes available, the synchronization module automatically uploads pending records to the central cloud server and removes successfully synchronized local copies.

This architecture eliminates dependency on internet connectivity while ensuring security, speed, and scalability.

5. Technical Architecture

The proposed solution consists of four major components:

1. **Mobile Application Layer**

Responsible for user interaction, camera access, authentication workflow, and local storage management.

2. **AI Inference Layer**

Performs face detection, liveness verification, and facial recognition using optimized lightweight models.

3. **Local Data Layer**

Stores biometric templates and authentication logs securely using SQLite.

4. **Cloud Synchronization Layer**

Uploads offline authentication records to AWS infrastructure once connectivity is restored.

6. System Architecture

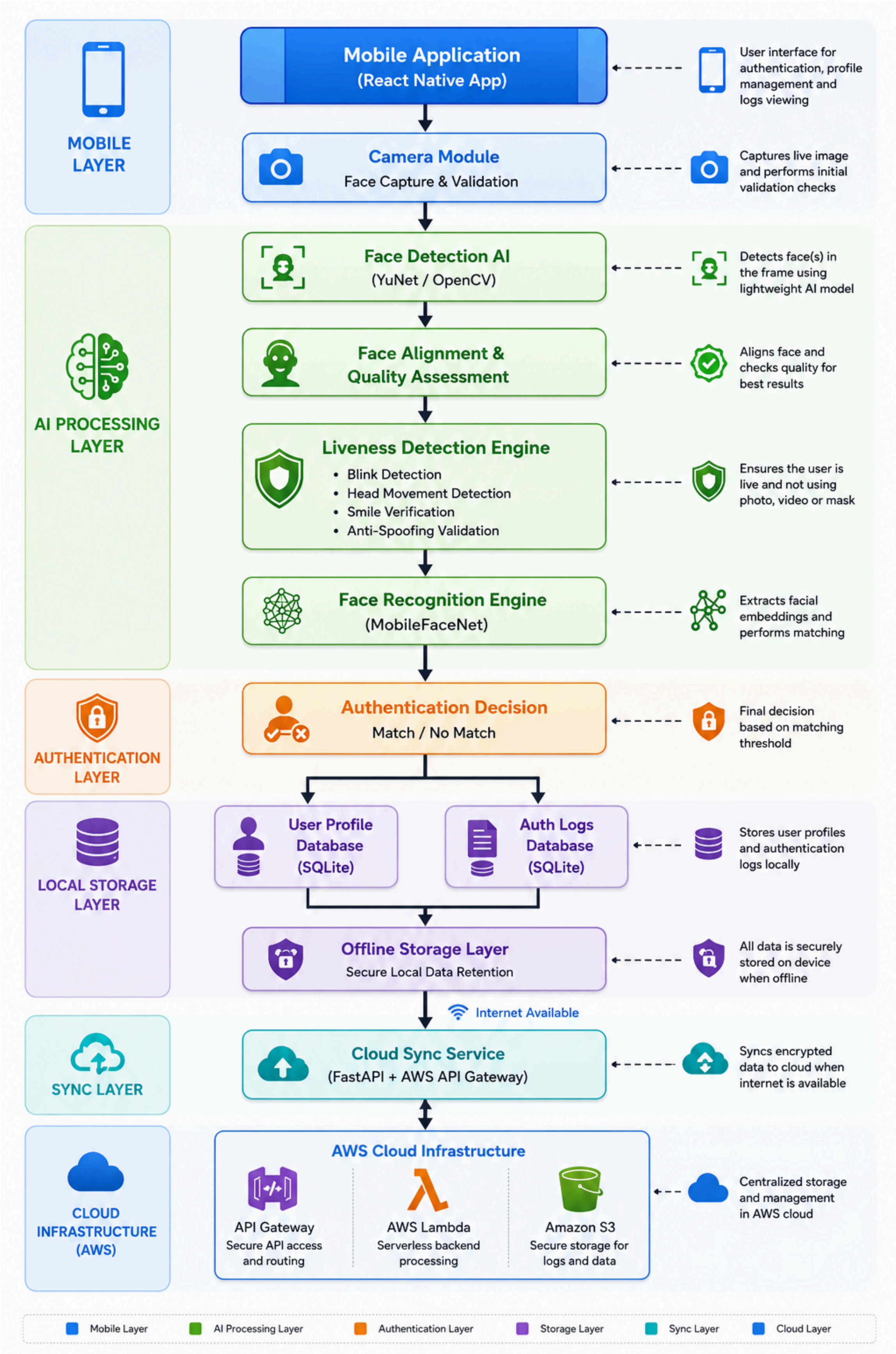
Authentication Workflow :-

Mobile Application
↓
User Initiates Authentication
↓
Image Capture Using Mobile Camera
↓
Face Detection
↓
Face Alignment and Quality Assessment
↓
Offline Liveness Detection
↓
Facial Feature Extraction
↓
Face Matching and Verification
↓
Authentication Decision
↓
Encrypted Local Storage
↓
Cloud Synchronization When Network Is Available
AWS Cloud Infrastructure

LivenessAuth AI performs facial recognition and liveness verification entirely on-device using lightweight edge-AI models.

Authentication records are securely stored locally and synchronized with cloud infrastructure when network connectivity becomes available, ensuring uninterrupted operations in zero-network environments.

Workflow Architecture



7. Technology Stack

Mobile Development
• React Native
• TypeScript

Artificial Intelligence & Computer Vision
• OpenCV
• MediaPipe
• MobileFaceNet
• MiniFASNet
• ONNX Runtime
• NumPy

Backend Services
• Python
• FastAPI

Database
• SQLite

Cloud Infrastructure
• AWS API Gateway
• AWS Lambda
• AWS S3

Version Control
• Git
• GitHub

8. Key Features

1. **Fully Offline Authentication**

Performs user authentication without requiring internet connectivity, ensuring uninterrupted operations in remote environments.

2. **Lightweight AI Models**

Optimized machine learning models designed specifically for mobile deployment with a total model size below 20 MB.

3. **Real-Time Performance**

Authentication process completed within one second on standard mid-range mobile devices.

4. **Multi-Layer Liveness Detection**

Protects against spoofing attacks through blink detection, facial movement analysis, smile verification, and texture-based anti-spoofing.

5. **Secure Local Storage**

Sensitive biometric information is securely stored and protected through encryption mechanisms.

6. **Cross-Platform Compatibility**

Supports Android and iOS devices through a unified React Native architecture.

7. **Automatic Sync and Purge**

Authentication logs are automatically synchronized with cloud infrastructure and removed from local storage after successful upload.

8. **Scalability**

Designed to support large-scale deployment across multiple operational regions and thousands of users.

9. Innovation and Uniqueness

LivenessAuth AI introduces a comprehensive edge-AI authentication framework specifically designed for low-connectivity and zero-network environments.

Unlike traditional cloud-dependent systems, the proposed solution executes the complete authentication pipeline directly on the device, ensuring reliability and security regardless of network availability.

Key innovations include:-

• Fully offline authentication workflow.
• Lightweight edge-AI architecture.
• Multi-layer anti-spoofing framework.
• Intelligent synchronization and purge mechanism.
• Mobile-optimized inference engine.
• Privacy-preserving local processing architecture.

10. Security Framework

Security is a core design principle of LivenessAuth AI.

The solution incorporates multiple security measures including:

• Encrypted biometric template storage.
• Secure local database management.
• Offline authentication processing.
• Multi-layer liveness verification.
• Spoofing attack prevention.
• Secure cloud synchronization.

• Audit logging and traceability.

These measures ensure the confidentiality, integrity, and authenticity of authentication operations.

11. Implementation Roadmap

Phase 1 – Requirement Analysis and Architecture Design
Phase 2 – AI Model Integration and Optimization
Phase 3 – Mobile Application Development
Phase 4 – Offline Authentication Implementation
Phase 5 – Sync and Purge Module Development
Phase 6 – Performance Testing and Optimization
Phase 7 – Documentation and Deployment

12. Expected Performance Metrics

Parameter	**Target**
Face Recognition Accuracy	Greater than 95%
Liveness Detection Accuracy	Greater than 95%
Authentication Time	Less than 1 Second
Total Model Size	Less than 20 MB
RAM Requirement	Minimum 3 GB
Android Support	Android 8.0 and Above
iOS Support	iOS 12 and Above
Offline Availability	100%

13. Expected Benefits

The proposed solution will provide significant operational advantages including:

• Reliable authentication in remote locations.
• Elimination of network dependency.
• Reduced authentication failures.
• Enhanced security through liveness detection.
• Improved operational efficiency.
• Lower infrastructure costs.
• Better user experience.
• Seamless integration with existing systems.

14. Future Enhancements

Future versions of LivenessAuth AI may include:

• Advanced deepfake detection.
• Aadhaar-based identity verification.
• Multi-factor biometric authentication.
• Geofencing-based access control.
• AI-powered anomaly detection.
• Centralized monitoring dashboard.
• Federated learning-based model updates.

15. Conclusion

LivenessAuth AI presents a practical, scalable, and innovative solution for secure identity verification in remote and zero-network environments. By leveraging lightweight edge-AI technologies, the proposed system delivers high-performance offline authentication while maintaining security, privacy, and operational reliability.

The solution directly addresses the objectives of NHAI Innovation Hackathon 7.0 and demonstrates a sustainable approach toward enabling secure digital authentication for field personnel operating in challenging connectivity conditions.

---

16. About the Participant

Anil Kushwaha

- **2.6+ years** of experience in Python Development and AI/ML application.
- Experience in Python, FastAPI, Artificial Intelligence, Machine Learning, Generative AI, LangChain, Retrieval-Augmented Generation (RAG), REST APIs, and backend development. and Backend Development.
- Developed offline AI assistants and AI-powered automation solutions.
- Strong expertise in designing lightweight AI applications for resource-constrained environments.
- **Proven track record** of delivering impactful projects, including **Offline AI Assistants** (ChatGPT-like) and Offline RAG-based PDF Question Answering Systems (Chatbot) without internet or external API dependency for Government projects at the Ministry of Defence, New Delhi.
- Worked on **Process Monitoring Agent** for **real-time system tracking** (CPU, Memory, Storage) and scalable backend services.

His experience in building lightweight AI solutions and deploying them in resource-constrained environments aligns closely with the objectives of the NHAI Innovation Hackathon 7.0.

Through this project, he aims to leverage edge AI technologies to create a secure, reliable, and efficient offline authentication system that can operate seamlessly in remote and zero-network environments.